Hill Sudani

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Summary

Computer Science student at Arizona State University (GPA 3.83) building rigorously validated ML and quantitative research systems – reproducing published interpretability research and stress-testing trading strategies with statistical rigor most student projects skip. Seeking Summer 2027 software engineering or ML/quantitative research internships.

Education

Arizona State University, Tempe, AZ — May 2028

Bachelor of Science in Computer Science; GPA: 3.83, Dean's List

Relevant Coursework: Data Structures & Algorithms, Computer Organization & Assembly, Linear Algebra, Probability & Statistics, Programming Languages

Experience

Freelance Software Developer | Shreedhar Group (Real Estate) — April 2025 – May 2025

Remote

- Shipped a production Next.js website with an integrated Grok API chatbot, live at shreedhargroup.vercel.app; achieved a **99** Google PageSpeed score.
- Building an internal automation platform: a lead pipeline (contact form to CRM to WhatsApp) via a self-hosted N8N instance on Oracle Cloud, secured behind a rate-limiting Vercel middleware layer, plus a Claude API-driven lead qualification system.

Projects

Code and full write-ups for every project are public on GitHub (link above).

Induction Heads: Causal Interpretability of a Small Transformer — May 2026

Python, PyTorch | github.com/Hill-Sudani/induction-heads

- Reproduced a published mechanistic-interpretability result (induction-head circuits): a 2-layer transformer reached **99.98%** task accuracy vs. **25.62%** for a 1-layer control, confirming the K-composition requirement.
- Proved causal, not just correlational, importance via targeted ablation – accuracy collapsed to **0.09%** with induction heads removed vs. a **0.66-point** drop for the cleanest control head – and pre-registered an OOD generalization threshold rather than adjusting it post-hoc.
- Found partial redundancy among induction heads: ablating 3 of 4 dropped accuracy to **65.8%**, but removing all four collapsed it to **0.09%**, showing the heads partially back each other up.

Adaptive Statistical Arbitrage with a Kalman Filter — July 2026

Python, statsmodels, scikit-learn | github.com/Hill-Sudani/stat-arb-kalman

- Built a leakage-aware pairs-trading pipeline – cointegration screening, walk-forward validated backtesting, and a Kalman-filtered adaptive hedge ratio – benchmarked against a static-OLS baseline and an MLP challenger.
- Stress-tested the strategy with deflated Sharpe ratios and 2,000-iteration Monte Carlo permutation testing, which correctly rejected an apparently profitable backtest (**11%** deflated-Sharpe probability post-correction) – a result a naive Sharpe calculation would have missed.
- Reported the adaptive Kalman hedge ratio underperforming the simpler static baseline (**-0.97%** vs. **+1.09%** net return) as-is, rather than discarding the less flattering result.

CUDA-Accelerated Matrix Multiplication — February 2026

C++, CUDA, Nvidia Nsight | github.com/Hill-Sudani/cuda-matmul

- Implemented and profiled naive and shared-memory-tiled CUDA kernels, reaching **14.4%** of cuBLAS throughput; used a roofline analysis to classify both kernels as memory-bound from Nsight-measured FLOP/s and DRAM bandwidth.
- Integrated the kernel into a neural network's forward pass and used the result (a **6x** slowdown from PCIe transfer overhead at small batch sizes) to correctly diagnose when GPU acceleration helps versus hurts.

NumPy Neural Network from Scratch (MNIST) — December 2025

Python, NumPy | github.com/Hill-Sudani/nn-from-scratch

- Implemented forward propagation and backpropagation manually (no PyTorch/TensorFlow), training a multi-layer network to **97.01%** MNIST test accuracy, verified against finite-difference gradient checks.
- Built a 784-128-64-10 architecture with He initialization and mini-batch gradient descent, validated by **9** unit tests including analytic-vs-numerical gradient comparison.

Technical Skills

Languages: Java, Python, C/C++, CUDA, JavaScript

Machine Learning: PyTorch, NumPy, Pandas, scikit-learn, Neural Networks & Backpropagation, Transformers & Attention, Mechanistic Interpretability, Classical ML (regression, classification, ensembles)

Quantitative/Statistical: Cointegration Testing, Kalman Filtering, Walk-Forward Validation, Deflated Sharpe Ratio, Monte Carlo Permutation Testing

Systems/Cloud: Git, N8N, Vercel, Oracle Cloud, AWS EC2, Nvidia Nsight